HANIKA ANGLICAN INTEGRATED POLYTECHNIC (HAIP) · Data Mining and Data Warehousing · FA1 Group Work

Dashboard Guide

Building and using the three Power BI dashboards

Project: University Library Analytics Data Warehouse

Group: **Group X** (replace with your group number)

Members: Member 1 (Data Architect) · Member 2 (ETL Specialist) · Member 3 (Analytics & Security Lead)

Database: SQLite 3.53.1 · ETL: Python 3 + pandas · Dashboards: Power BI Desktop

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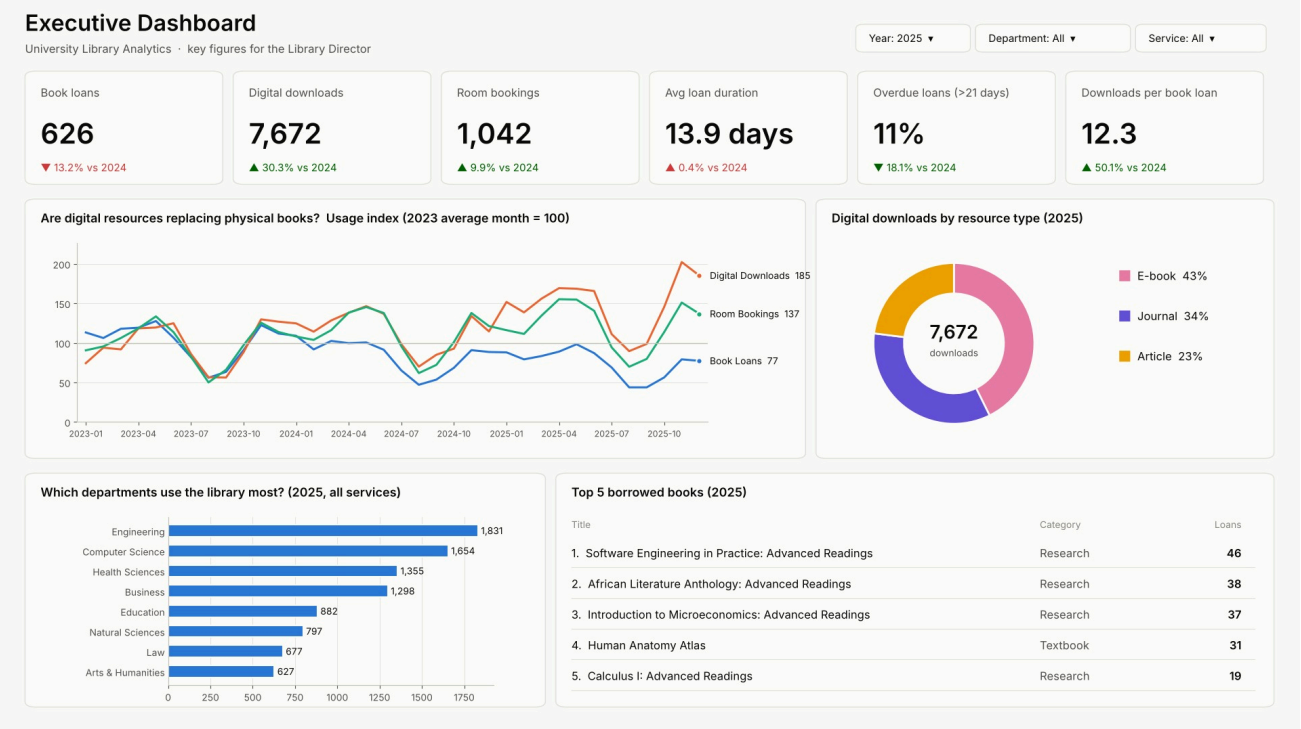
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1. The three dashboards

Dashboard	For	Answers
Executive	Library Director	headline KPIs vs last year, digital vs physical trend, department ranking, resource mix, top books
Department	Deans, department heads	usage per department and service, pivot of resource types, loan duration, undergraduate vs graduate
Operational	Desk staff, room managers	last 30 days, peak-time heat map, daily bookings, room utilisation, books not returned, alerts

The images in this folder (`executive_dashboard.jpg`, `department_dashboard.jpg`, `operational_dashboard.jpg`) are the **design** of each page, drawn from the real warehouse data by `build_dashboard_images.py`. Section 4 shows how to build the same pages in Power BI; after building, save your own screenshots over these files.



Executive dashboard (design)

2. What you need

Item	Where	Notes
Power BI Desktop	free from Microsoft (Microsoft Store or powerbi.microsoft.com)	Windows only
The data	`06_Dashboards/PowerBI_Data/*.csv` **or** `03_Database_Files/library_dw.db`	CSV = easiest
DAX measures	`06_Dashboards/PowerBI_DAX_measures.txt`	copy & paste
Theme	`06_Dashboards/LibraryDW_PowerBI_theme.json`	colours used in the designs
SQLite ODBC driver (only for option B)	"SQLite ODBC Driver" by Christian Werner (ch-werner.de/sqliteodbc), **64-bit** version	free

3. Load the data and build the model

3.1 Option A (recommended): the CSV files

1. Run the ETL once (`python etl_script.py`) - it writes fresh CSV files into `06_Dashboards/PowerBI_Data`.
2. Power BI Desktop → **Home → Get data → Text/CSV** → choose `dim_date.csv` → **Transform Data**.
3. Repeat for the other 11 table files (`dim_*.csv`, `fact_*.csv`). Tip: after the first file, use **New Source → Text/CSV** inside Power Query.
4. `dim_student.csv` contains only the **masked** student ID - the real ID never reaches Power BI.

3.2 Option B: live connection to library_dw.db (ODBC)

1. Install the 64-bit SQLite ODBC driver.
2. Windows → **ODBC Data Sources (64-bit)** → tab *User DSN* → **Add** → *SQLite3 ODBC Driver* → Data Source Name `LibraryDW`, Database Name: browse to `library_dw.db` → OK.
3. Power BI → **Get data → More... → Other → ODBC** → DSN `LibraryDW` → Navigator: tick the 3 `fact_*` tables, the dimensions **and** `v_dim_student_masked` **instead of** `dim_student` (privacy) → **Transform Data**.
4. In Power Query rename `v_dim_student_masked` to `dim_student`.

3.3 Clean-up in Power Query (both options)

Table	Step
dim_date	filter `date_key` → untick **-1** (the "not returned" row; a date table must have continuous dates); set `full_date` and `week_start_date` to type **Date**
all tables	check that keys are *Whole number*, `duration_hours` *Decimal number*, flags (`is_returned`, `is_overdue`, `is_walk_in` ...) *Whole number*

Click **Close & Apply**.

3.4 Relationships (Model view)

Power BI usually detects them; check this list and fix if needed (all **many-to-one, single direction**, from the dimension to the fact):

From (one)	To (many)	Active
dim_date[date_key]	fact_book_loan[checkout_date_key]	yes
dim_date[date_key]	fact_book_loan[return_date_key]	**no** (inactive; use with USERELATIONSHIP if needed)
dim_date[date_key]	fact_digital_usage[date_key]	yes
dim_date[date_key]	fact_room_booking[date_key]	yes
dim_department[department_key]	each of the 3 facts [department_key]	yes
dim_student[student_key]	fact_book_loan, fact_room_booking [student_key]	yes
dim_book[book_key]	fact_book_loan[book_key]	yes
dim_resource[resource_key]	fact_digital_usage[resource_key]	yes
dim_user_type[user_type_key]	fact_digital_usage[user_type_key]	yes

From (one)	To (many)	Active
dim_room / dim_time_slot / dim_booking_purpose	fact_room_booking	yes

The result is the **galaxy schema** of the design document: three stars sharing `dim_date` and `dim_department`.

3.5 Date table, sorting and the hierarchy

1. Select `dim_date` → **Table tools → Mark as date table** → column `full_date`.
2. Select `month_name` → **Column tools → Sort by column → month_num**. Do the same: `month_short` by `month_num`, `day_name` by `day_of_week`, `slot_label` (in dim_time_slot) by `start_hour`.
3. Create the hierarchy for drill-down: in the Data pane right-click `year_num` → **Create hierarchy**; then right-click `quarter_name`, `month_name`, `day_of_month` → **Add to hierarchy** in that order. Name it **Calendar**.
4. Hide the key columns (right-click → Hide in report view) so users only see readable fields.

3.6 Measures and theme

1. **Home → Enter data** → name the table `_Measures` → Load.
2. Select `_Measures` → **New measure** → paste one measure from `PowerBI_DAX_measures.txt` → Enter. Repeat for every measure (Total Loans, Total Downloads, Total Bookings, Total Usage, Avg Loan Days, Overdue %, YoY measures, Index measures, Room Utilisation %, Walk-in %, Books Currently Out, Journal Share %, Anomaly).
3. Format: `%` measures → *Percentage*, 1 decimal; totals → *Whole number* with thousands separator.
4. **View → Themes → Browse for themes** → `LibraryDW_PowerBI_theme.json`.

One extra **calculated column** for the Operational page (select `fact_book_loan` → **New column**):

```
Days Out =
IF ( fact_book_loan[is_returned] = 0,
     DATEDIFF ( RELATED ( dim_date[full_date] ), DATE ( 2025, 12, 31 ), DAY ) )

Out Bucket =
SWITCH ( TRUE (),
    ISBLANK ( fact_book_loan[Days Out] ), BLANK (),
    fact_book_loan[Days Out] <= 14, "0-14 d",
    fact_book_loan[Days Out] <= 30, "15-30 d",
    fact_book_loan[Days Out] <= 90, "31-90 d",
    "90+ d" )
```

(In daily use replace `DATE ( 2025, 12, 31 )` by `TODAY ()`.)

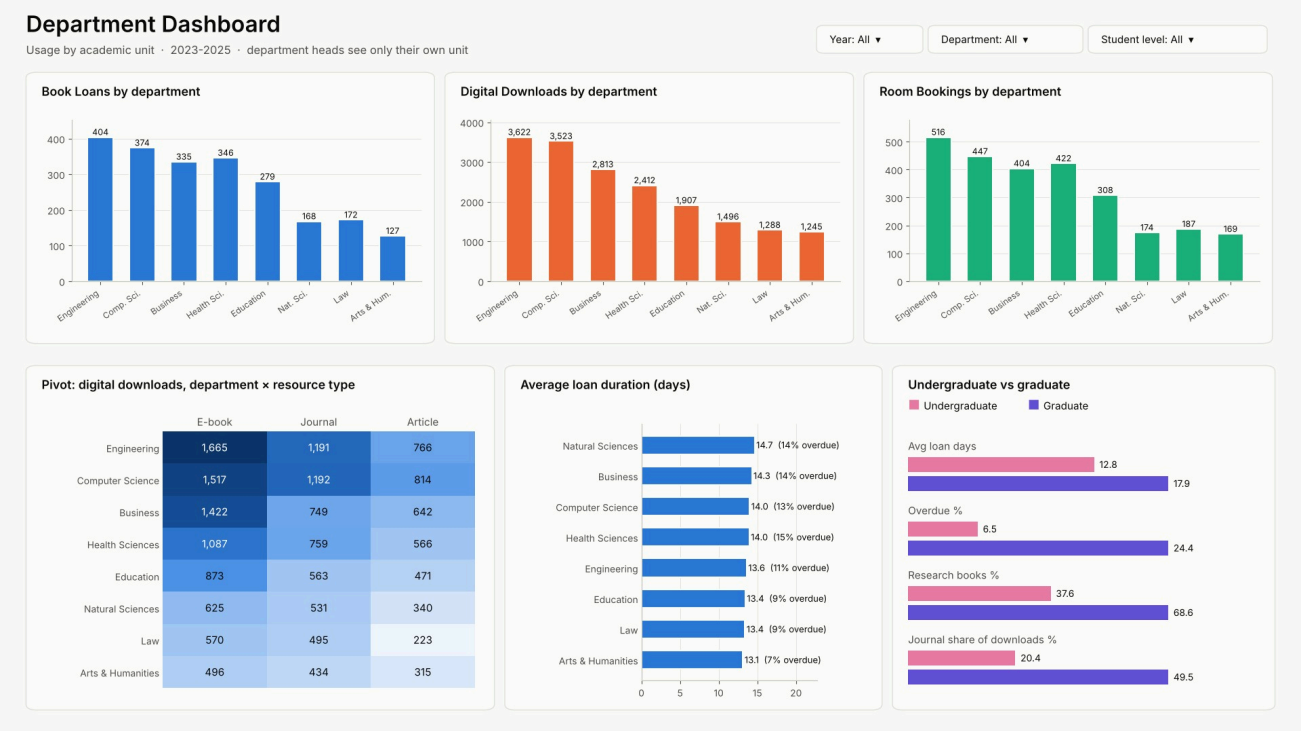
4. Build the pages

Canvas: **View → Page view → Fit to page**, page size 16:9. Rename pages *Executive*, *Department*, *Operational*.

4.1 Executive page

#	Visual	Fields	Settings
1	Slicer (Dropdown)	dim_date[year_num]	select 2025
2	Slicer (Dropdown)	dim_department[department_name]	
3	6 × Card	Total Loans, Total Downloads, Total Bookings, Avg Loan Days, Overdue %, Downloads per Loan	under each card a small card with the YoY % measure (or, in *Card (new)*, add the YoY measure as **reference label**)
4	Line chart	X: dim_date[month_label]; Y: Loans Index, Downloads Index, Bookings Index	**one axis** (indexes are comparable); edit interactions so the Year slicer does not filter it
5	Donut chart	Legend: dim_resource[resource_type]; Values: Total Downloads	detail labels: category + percent
6	Bar chart (horizontal)	Y: dim_department[department_name]; X: Total Usage	sort descending; filter department ≠ Unknown
7	Table	dim_book[title], dim_book[book_category], Total Loans	filter *title*: **Top N = 5 by Total Loans**; exclude "Unknown Book"

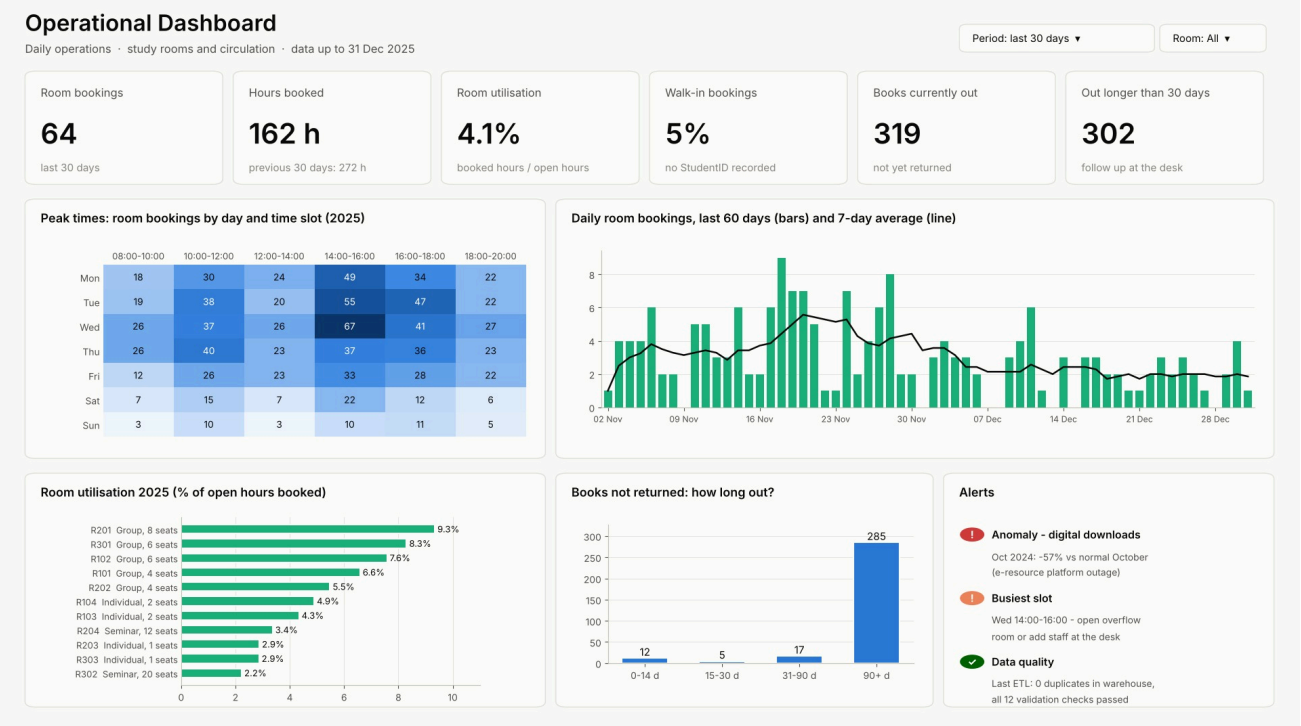
4.2 Department page



#	Visual	Fields	Settings
1	Slicers	year_num, department_name, dim_student[student_level]	
2	3 × Clustered column chart	X: department_name; Y: Total Loans / Total Downloads / Total Bookings	one chart per service, each in its service colour (blue / orange / green)
3	**Matrix** (the pivot)	Rows: department_name; Columns: dim_resource[resource_type]; Values: Total Downloads	Format → Cell elements → **Background colour** → gradient (heat map)
4	Bar chart	Y: department_name; X: Avg Loan Days; Tooltips: Overdue %	data labels on

#	Visual	Fields	Settings
5	Clustered bar chart	Y: dim_student[student_level]; X: Avg Loan Days, Overdue % (two small charts)	filter level ≠ Unknown
6	Clustered bar chart	Y: dim_user_type[student_level]; X: Journal Share %	shows UG vs Grad for digital

4.3 Operational page



#	Visual	Fields	Settings
1	Slicer (Between)	dim_date[full_date]	set to the last 30 days (or Relative date: *Last 30 days* in daily use)
2	Cards	Total Bookings, Hours Booked, Room Utilisation %, Walk-in %, Books Currently Out	Books Currently Out: edit interactions → not filtered by the date slicer
3	**Matrix heat map**	Rows: dim_date[day_name]; Columns: dim_time_slot[slot_label]; Values: Total Bookings	background colour gradient; filter slot ≠ Unknown
4	Line and clustered column chart	X: dim_date[full_date]; Column: Total Bookings	last 60 days; add a trend/average line in Analytics
5	Bar chart	Y: dim_room[room_number]; X: Room Utilisation %; Tooltips: room_type, capacity, has_projector	sorted descending
6	Column chart	X: fact_book_loan[Out Bucket]; Y: count of loan_key	
7	Table "Alerts"	dim_date[month_label], Total Downloads, Downloads Normal Month, Anomaly	visual filter: Anomaly = "ANOMALY - sudden drop"

4.4 Save and export

- **File → Save as** `LibraryDW_Dashboards.pbix` in `06_Dashboards`.
- For the submission images: show each page full screen and save a screenshot as `executive_dashboard.jpg`, `department_dashboard.jpg`, `operational_dashboard.jpg` (or **File → Export → Export to PDF**).

5. OLAP operations in Power BI

Operation	How to do it in the report
Drill-down	put the *Calendar* hierarchy on the X axis of a chart → click the **down arrow** (drill mode) → click a year bar → its quarters → a quarter → its months → days
Roll-up	click the **up arrow** (drill up), or the double down arrow "*Go to the next level*" to see all quarters together
Slice	one slicer, e.g. Year = 2024 and Quarter = Q1
Dice	several slicers together, e.g. Department = Engineering + Month = March + Resource type = Journal
Pivot	the Matrix visual: drag a field from *Rows* to *Columns* to rotate the view

6. Department-level security in Power BI (Row-Level Security)

When the report is published to the Power BI service, department heads must see only their own department - the same rule as `secure_access.py`.

1. Load `PowerBI_Data/powerbi_rls_users.csv` (e-mail → department code) as table `rls_users` (no relationship needed).
2. **Modeling → Manage roles → New role** `Department Head` → table `dim_department` → DAX filter:

```
[department_code] =
    LOOKUPVALUE ( rls_users[department_code], rls_users[user_email], USERPRINCIPALNAME () )
```

1. Because `dim_department` filters all three facts, every visual now shows only that department.
2. Test: **Modeling → View as → Department Head** + *Other user* `head.cs@university.example` → only Computer Science is visible.
3. In the Power BI service: dataset → Security → add the department heads' accounts to the role. The Director and analysts are not added to the role and see everything.

7. Refreshing the data

1. The nightly ETL (00:30) reloads the warehouse and rewrites the CSV files.
2. In Power BI Desktop: **Home → Refresh**. In the Power BI service: schedule a refresh at 06:00 (needs the *on-premises data gateway* on the computer that holds the files).
3. Always check the *Data quality* alert on the Operational page after a refresh.

8. Using the dashboards (for everyone)

- **Hover** over any bar, line point or cell to see the exact numbers.
- **Click** a bar (e.g. Engineering) to filter all other visuals on the page; click again to clear.
- **Slicers** at the top change the whole page; *Ctrl+click* to choose several values.
- **Drill**: on charts with the Calendar hierarchy, use the arrows in the top-right corner of the visual.

- **See the data behind a visual**: right-click → *Show as a table*; **export** with … → *Export data* (staff accounts only).
- **Reset** to the default view: *Reset to default* (top bar in the Power BI service).
- KPI arrows: ▲/▼ compare with the same period last year; green means good for the library, red means needs attention (e.g. a ▼ in loans is red, a ▼ in overdue loans is green).